Sec: Sr. IIT | **Race -22** | **Date: 08-03-24**

PHYSICS **Max Marks: 60**

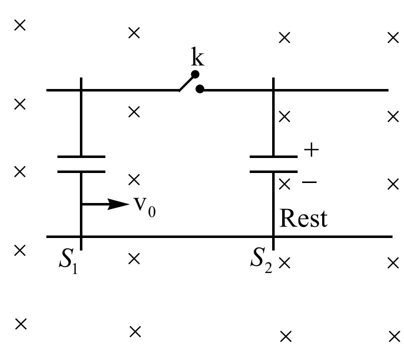
SECTION-1

(ONE OR MORE OPTIONS CORRECT TYPE)

This section contains 10 multiple choice equations. Each question has four choices (**A**) (**B**),(**C**) and (**D**) out of which **ONE** or **MORE THAN ONE** are correct.

Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases.

1. Two long straight horizontal rails are fixed in space where uniform magnetic field of intensity 2 Tesla is present. Two sliders S_1 and S_2, are containing identical capacitors each of capacitance $C = 2\times10^{-2}\,F$ are kept on rails as shown. Mass of slider S_1 and S_2 are 0.50 kg and 0.25 kg respectively. Initially capacitor in slider S_2 is charged to 40 V while in S_1 is uncharged. Initially slider S_1 is moving uniformly at speed $v_0 = 6\,m/s$, and slider S_2 is at rest as shown in figure. The distance between parallel rails is 50 cm.



At t = 0 key k is closed. Finally the two sliders attains steady state motion. Ignore friction and resistance in sliders and in rails also.

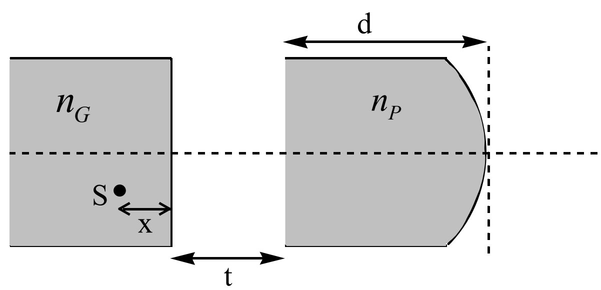
A) Find speed of slider $S_1 = 2\ m/s$

B) Find speed of slider $S_1 = 4\ m/s$

C) Total heat released in process is 8 J

D) Total heat released in process is 11 J

2. As shown below, a point light source (S) is inside a glass block of refractive index $n_G = \frac{3}{2}$, at a distance y = 2 mm from the optical axis and distance x = 3 cm from the flat surface. A thick plano-convex lens made of transparent plastic material of refractive index $n_P = 2$, is placed near the source as shown. The thickness of the lens is d, and the radius of convex surface is R = 15 cm. The width of air gap between the glass block and the lens is 't'. The width of air gap t and thickness of lens d is adjusted so that light (paraxial rays) from point source forms always a parallel beam after passing through the lens. Select **CORRECT** option(s).



A) For d = 16cm, width of air gap t = 5 cm

B) For thickness d = 10cm, width of air gap t = 8 cm

C) The maximum adjustable value of d = 26 cm

D) The maximum adjustable value of d = 16 cm

3. Light of wavelength $\lambda = 248$ nm is incident on an imaginary photoelectric tube. Its current-voltage characteristic is shown in figure-1, it consists of two straight lines as shown in figure-1. The emitter and the collector plates are made of same, imaginary metal has work function W = 1.5 eV. A parallel combination of two such identical photo electric tubes is getting used as solar cell to supply current to an external resistance R as shown in figure-2.

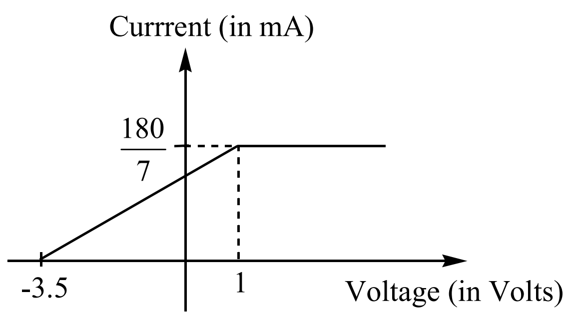


Figure-1

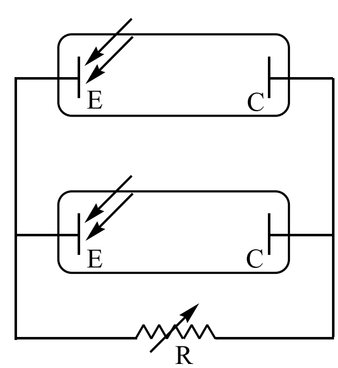


Figure-2

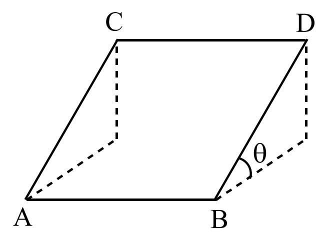
A) The maximum power that can be obtained from this parallel combination of photo electric tube working as a solar cell is $7\times10^{-2}W$

B) The maximum power that can be obtained from this parallel combination of photo electric tube working as a solar cell is $3.5\times10^{-2}W$

C) Resistance R satisfying maximum power condition of solar cell described in figure-2 is 175Ω

D) Resistance R satisfying maximum power condition of solar cell described in figure-2 is 87.5Ω

4. According to quantum mechanics, a particle of momentum P can be regarded as a plane matter wave with wavelength $\lambda=\frac{h}{P}$, where h is plank constant. As shown in figure, ABDC is a flat square of side length L at an inclined angle $\theta=30°$ to the horizontal plane. A neutron beam of kinetic energy E_0 is divided into two beam at point A. One beam moves along path ACD and other beam along the path ABD. When the two beam meet at point D they interfere. The mass of neutron is m. Gravitational acceleration is g.



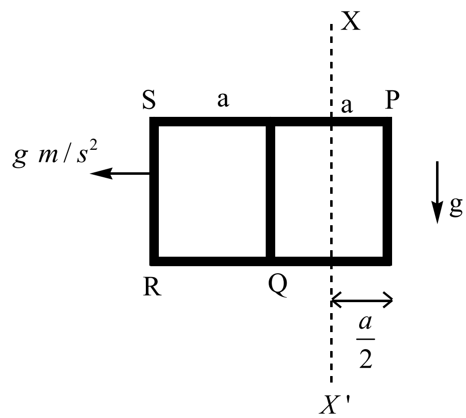
A) At point D matter wave following path ABD leads in phase over other one following path ACD

B) At point D matter wave following path ACD leads in phase over one following path ABD

C) $\sqrt{2mE_0}-\sqrt{m\left(2E_0-mgL\right)}-\frac{h}{2L}=0$; is a condition which satisfies interference minima at D

D) $\sqrt{2mE_0}-\sqrt{m\left(2E_0-mgL\right)}-\frac{h}{2L}=0$; is a condition which satisfies interference maxima at point D

5. A conducting loop is shown figure, it is made of seven identical wires each of length a, resistivity ρ, area of cross-section S. The frame lies in a vertical plane. XX' is an imaginary boundary separating space in two parts. Right to XX', a uniform gravitational field $\vec{g}$ exists (figure) whereas no gravitational field is present left of XX'. Instantaneously frame is accelerating in leftwards with acceleration $g\,m/s^2$. e and m is electronic charge and mass respectively. Ignore inductive and capacitive properties of loop. Instantaneously



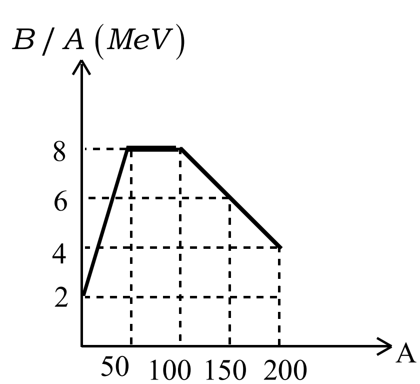
A) The potential difference between points P and Q is $\frac{4mga}{15\,e}$

B) The potential difference between points P and Q is $\frac{8}{15}\frac{mga}{e}$

C) Current in branch RS is $\frac{mgS}{15\rho e}$

D) Current in branch RS is $\frac{4mgS}{15\rho e}$

6. If the graph of Binding energy (B/A) per nucleon versus mass number (A) looks like as shown in figure. Then using the curve choose **CORRECT** option(s)



A) Fusion of two nuclei with mass numbers 30 and 45 into one nucleons of mass number 75 will release energy

B) Fission of nuclei with mass number 80 into two nuclei of equal mass number will have Q value 256 MeV.

C) Fission of nuclei with mass number 150 into two nuclei of equal number will release energy.

D) Fusion of two nuclei with mass number 10 and 20 in to a nucleus of mass number 30 will release energy

7. A stationary sound source emits a wave that is received by a stationary observer as $P = P_0 \sin(kx - \omega t)$. Now

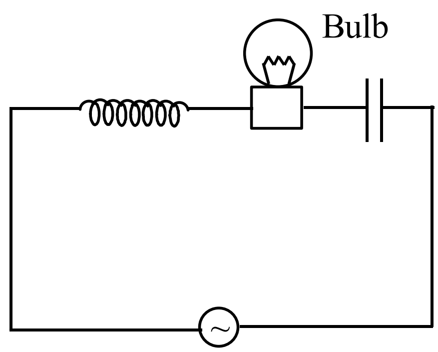
A) If source moves towards observer, observer will perceive higher k as well as ω.

B) If source moves towards observer, observer will perceive lower k but higher ω

C) If observer moves towards source, observer will perceive same k but higher ω

D) If observer moves towards source, observer will perceive higher k as well as ω

8. In the circuit shown, the inductor and capacitor are ideal and source has no internal resistance. The circuit is in resonance. Discuss about the new steady state current obtained in circuit after making changes explained in options.



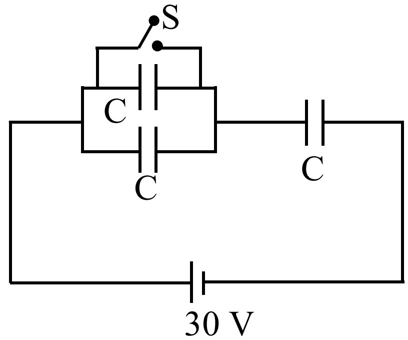
A) If a dielectric slab is inserted in the capacitor, current will now lag behind the applied voltage.

B) If an iron rod is inserted in the inductor, current will now lead the applied voltage.

C) If an iron rod is inserted in the inductor, current will now lag behind the applied voltage,

D) If a dielectric slab is inserted in the capacitor current will now lead the applied voltage.

9. Three capacitors each having capacitance $C = 2\ \mu F$ are connected with a battery of emf 30 V as shown in the figure. On closing switch S circuit acquires new steady state



A) The amount of charge flown through the battery is 20 μC

B) The heat generated in the circuit is 0.6 mJ

C) The energy supplied by the battery is 0.6 mJ

D) The amount of charge flown through the switch S is 60 μC

10. An X-ray tube operating at 84 kV & 10 mA. Only 1% of the electric power supplied is converted into X-rays. If target has a mass of 300 gm & specific heat of $0.04\ cal/gm/°C$. Then choose the **CORRECT** statement(s) (use 1 cal = 4.2 J & hc = 12400 eV-$\overset{0}{A}$) Assume entire thermal energy is absorbed by the target.

A) Cut-off wavelength is approximately $0.14\ \overset{0}{A}$

B) Cut-off wavelength is approximately $6.8\ \overset{0}{A}$

C) Average rate of temperature rise of target (assuming no heat loss) is $16.5\ °C/s$

D) Average rate of temperature rise of target (assuming no heat loss) is $1.7\ °C/s$

SECTION-II

INTEGER TYPE

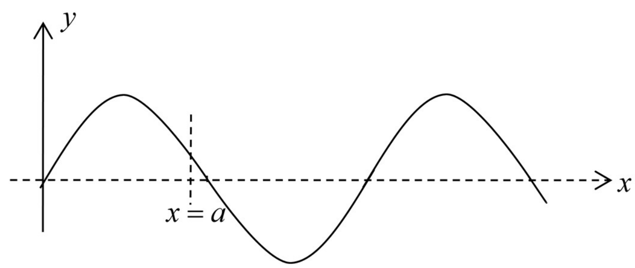
This section contains 10 questions. The answer to each question is a **single digit integer, ranging** from 0 to 9 (both inclusive).

Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases.

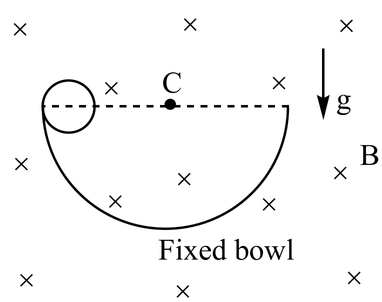
11. The figure shows a snap photograph of a vibrating string at time instant t = 1 sec. Instantaneous rate of energy transfer across the cross section at $x = a$ is half of it's maximum possible value. Wave energy in string at same instant between $x = 0$ to $x = a$ is P joule (approximately). Find P ?

Given: Progressive wave in string is sinusoidal. Linear mass density of string = 6 kg/m

Wave length = 2.2 m, Amplitude of wave = 5 mm, Angular frequency of wave = 200 rad/s.

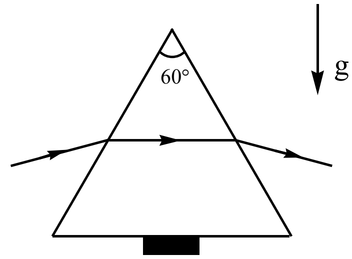


12. A solid sphere is held at rest on the fixed hemispherical bowl made of insulating material such that centre of sphere lies on the horizontal diameter as shown in figure. Total charge $q = 5.1\times10^{-1}\,C$ and mass m = 42 grams, both are distributed uniformly in the sphere. There exists uniform magnetic field B is space as shown in figure. Find limiting value of B (in Tesla) so that on releasing sphere, it do not leaves contact with bowl in course of motion. Consider contact surface to be sufficient rough. Spherical bowl is made of non-magnetic material. Radius of bowl = 10 cm, radius of sphere 3 cm. $g = 10\,m/s^2$.



13. An equilateral glass prism kept on a table has refractive index of $\mu = \sqrt{2}$. It is illuminated by a narrow laser beam having power P_0 and wavelength λ. The path of the laser beam inside the prism is parallel to the base of the prism. The change in reading of weight of the prism due to the incident laser beam is $\frac{P_0}{c}\left(\frac{\sqrt{x}-1}{\sqrt{y}}\right)$.

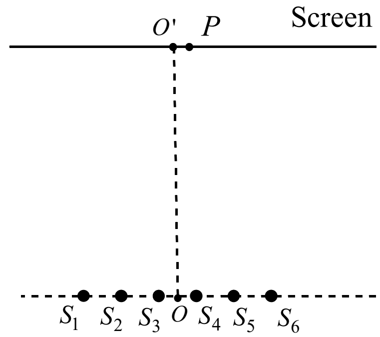
Find $x + y$? c is the speed of light. (Ignore any partial reflection)



14. In an optical bench experiment to determine the focal length of the concave mirror, the least count of the bench scale (meter scale) is 0.1 cm. The object needle reading is 10.7 cm and the image needle reading is 5.7 cm. The mirror is positioned at 15.7 cm. The maximum possible percentage error in the focal length of the mirror with this set of readings is $\frac{10}{x}$. Find x ?

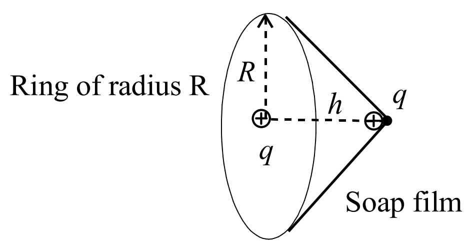
15. Consider the ground state energy of the hydrogen atom (a system consisting of a proton and an electron) in the non-relativistic model with the proton fixed at a point. If the electron is at infinite distance from the proton, the energy of the configuration is defined to be zero. Find the modulus of ground state energy in terms of dimensions units defined by putting the mass of the electron $m_e = 10$, reduced planck constant $\hbar = 10$ and $k_e e^2 = 10$. Where k_e is the Coulomb's constant and e denotes the electron's charge.

16. Light source $(S_1, S_2, S_3, S_4, S_5, S_6)$ are in a line and all are coherent light sources with same wavelength (λ). Distance between consecutive sources is 'a'. Sources S_1 and S_2 are oscillating in phase. All other are oscillating at a phase lag of $\frac{\pi}{2}$ with both S_1 and S_2. In wide central region of screen intensity due to all individual sources is same equal to I. 'O' is midpoint of $S_3\ S_4$. OO' is perpendicular to line of sources. On the screen at point P, exactly opposite to S_4, resultant intensity found is 2 I when only $S_3 \& S_4$ are working and nowhere else between O' and P intensity was found to be 2I. Now if all sources are working the resultant intensity at point P is $2x \cdot I$. Fill the value of x in OMR sheet.



17. A protonium atom consists of a proton and an antiproton. An antiproton is negatively charged particle, mass and modulus of change of which are equal to those of a proton. If binding energy of an electron in hydrogen atom is $\frac{40}{3}$ eV and mass of proton is 1800 times the mass of electron, find the energy of a photon emitted in transition of a protonium atom from the first excited state to ground state. Give your answer in multiples of 'keV'

18. Slant surface of cone (approximately) shown in figure is a soap film of surface tension T created on a non conducting fixed ring of radius R. At the centre of ring a charge +q is fixed and in apex region of cone charge +2q is fixed to the film which can be treated as point charge and free to move along with soap material. Material of soap solution is non conducting. System is in equilibrium. The relation $x\pi^2\varepsilon_0 Th^3 - q^2 = 0$ satisfies the equilibrium of charge in apex region. Find x ? (Assume h << R and also the space to be gravity free)



19. We have a thin non conducting spherical shell of radius R. One half of the sphere carrying surface change density σ_1 and rest half carrying σ_2. Magnitude of force of electrostatic interaction between hemispherical part carrying surface charge density σ_1 and rest of the hemispherical portion is $F_0 = \alpha \times 10^4$ Newton. Here α is an integer. Value of α will be ? (given $\sigma_1 = \sqrt{\varepsilon_0} \times 10^2\, C/m^2, \sigma_2 = 4\sigma_1, R = \sqrt{\frac{3}{\pi}}\, m$)

20. Consider a cell of emf E and internal resistance r. The current drawn (i) from cell can be varied. Output power from cell is same when current drawn is $i_1 = 2A$ and $i_2 = 8A$ .For what value of current (in A) the output power will be maximum?

CHEMISTRY **Max Marks: 60**

SECTION-1

(ONE OR MORE OPTIONS CORRECT TYPE)

This section contains 10 multiple choice equations. Each question has four choices (**A**) (**B**),(**C**) and (**D**) out of which **ONE** or **MORE THAN ONE** are correct.

Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases.

21. Select the correct statement(s):

A) A cubic system possesses a total of 23 elements of symmetry.

B) For triclinic system $a \neq b \neq c$ and $\alpha \neq \beta \neq \gamma \neq 90$.

C) Unit cell of c.c.p is f.c.c.

D) Monoclinic crystal system has body centred unit cell.

22. 1 lt of 1M $CuSO_4(aq)$ is electrolysed by passing 0.01 equivalent charge using inert electrode.

Which is/are correct about above process?

A) Intensity of blue colour changes during electrolysis

B) No change in colour intensity had Cu-electrode used

C) $\left[H^+\right] = 10^{-8}M$ after electrolysis

D) During electrolysis the libereated O_2 requires 28ml of CH_4 at 1 atm and 273 k for complete reaction.

23. Select the correct statement(s):

A) The rate law of the elementary reaction: $A + B \xrightarrow{k} C + D$

Must be : $\text{rate} = K[A]^1[B]^1$

B) The rate law for the complex reaction

$A + 2B \to C$, might be $r = K[A]^1[B]^1$

C) The partial orders may differ from the coefficients in the balanced reaction

D) Partial orders can never be zero or negative

24. Consider the following solution:

0.1m $RNH_3^+Cl^-(I)$; 0.1 m NaCl(II); 0.1m urea (III); 0.1m $NaKC_2O_4.10H_2O(IV)$ (assume complete dissociation)

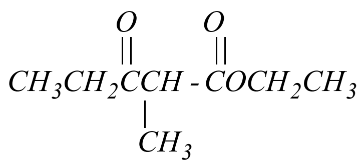
A) solution (IV) has highest boiling point

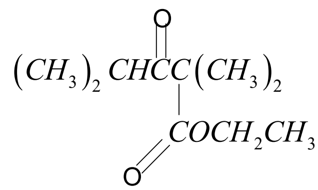
B) solution (III) has highest freezing point

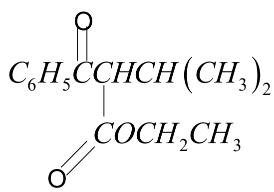
C) solution (I) and solution (II) have same osmotic pressure.

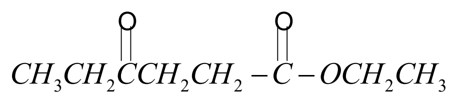
D) solution (III) has lowest osmotic pressure

25. Which of the following keto esters is/are not likely to have been prepared in good yield by a Claisen condensation?

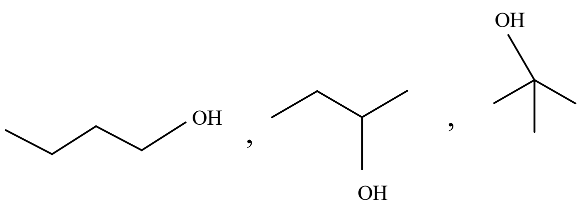
A) 

B) 

C) 

D) 

26. Lucas test is used to make distinction between



alcohols

$$ROH + \underset{conc.}{HCl} \xrightarrow{anhydrous\ ZnCl_2} \underset{White\ turbidity}{RCl \downarrow} + H_2O$$

True statements regarding test are

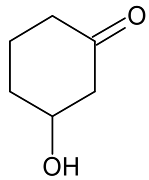
A) alcohols behave as Lewis base

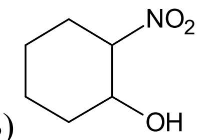
B) Greater the value of pK_a (of alcohol), greater is the reactivity with conc. HCl and thus faster the formation of white turbidity.

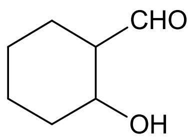
C) Alcohol which readily reacts with Na metal, will give turbidity readily

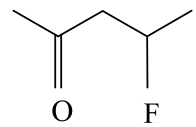
D) Alcohol which gives red colour in Victor Meyer test, will give turbidity at slower rate than those giving blue or white colour in Victor Meyer test.

27. The compounds which readily undergo E_{1cb} mechanism in basic medium are?

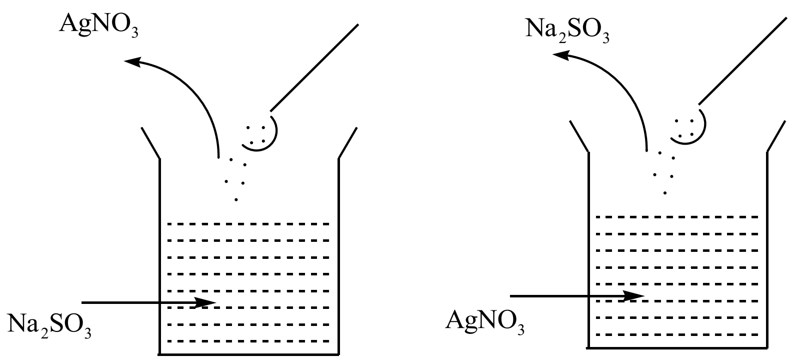
A) 

B) 

C) 

D) 

28. Observe the following experiments:



(I) (II)

The correct statement(s) about set up (I) and (II) is/are:

A) In set up (I) initially no precipitate is formed but on addition of excess of reagent and followed by boiling gives greyish black ppt.

B) In set up (II) initially a white precipitate is formed as well as on boiling gives grayish black ppt

C) In set up (I) the precipitate formed with excess $AgNO_3$ which is soluble ammonia solution.

D) In set up (II) the precipitate is soluble is dilute nitric acid and gives gas with suffocating smell of burning sulphur.

29. $Co^{+2}(aq) + SCN^{-}(aq) \longrightarrow complex(P)$

$Ni^{+2}(aq) + dim\,ethyl\,glyoxime \xrightarrow{NH_4OH} Complex(Q)$

Choose correct statement(s) regarding P & Q:

A) P is tetrathiocyanato – N- cobaltate (II) ion

B) Q is bis (di methylglyoximato) nickel (II)

C) P is tetrahedral & Q is square planar

D) P is paramagnetic with three unpaired electrons and Q is diamagnetic

30. The oxo acids of chlorine $HOCl, HClO_2, HClO_3 \& HClO_4$ differ in:

A) Hybridisation of chlorine

B) number of $P_\pi - d_\pi$ bonds

C) number of lone pairs of electrons on chlorine

D) magnetic nature.

SECTION-II

INTEGER TYPE

This section contains 10 questions. The answer to each question is a **single digit integer, ranging** from 0 to 9 (both inclusive).

Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases.

31. At an applied discharge potential, if the volume of O_2 and Cl_2 liberated by the electrolysis of 1 liter of an aqueous NaCl solution are in the 2:1. After consumption of 4.825 coulombs of charge, the pH of the solution is____________

32. In a solid AB having rock salt structure, A are atoms in FCC. If all the FCC atoms along one of the axes are removed. Then what will be the sum of atoms in compound AB?

33. The number of incorrect statement(s) among the following is:

I) out of $PO_4^{3-}, \left[Fe(CN)_6\right]^{4-}, SO_4^{2-}$ and $Cl^{\ominus}$ Flocculating value of $\left[Fe(CN)_6\right]^{4-}$ is maximum

II) In chemisorptions, adsorption of gas on a solid surface generally follows first order kinetics at high pressure.

III) An emulsion is a homogeneous system

IV) Stability of colloidal system consisting of micelles obtained from soap units is attributed to the negative charge present on micelles.

V) $Fe(OH)_3$, $Al(OH)_3$,, eosin and Sb_2S_3 are negatively charged sols.

34. Identify the mechanism corresponding to major product formed when 2-bromopentane is treated with following

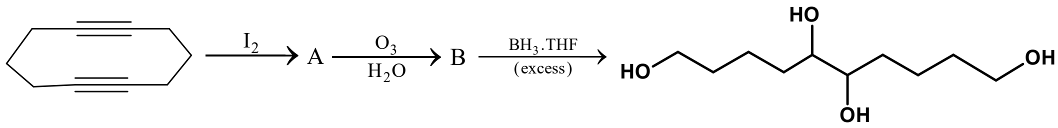
(A) NaOEt

(B) NaI/DMSO

(C) Sodium benzoate

and report your answer as sum of molecularities.

For example, if you identify it to be S_N^1, S_N^2, E^2 then you report 1+2+2=5

35. 

How many of the following statement(s) are correct regarding above sequence

1) Structure A is bicyclic

2) B Shows positive Tollen's test, positive 2,4-DNP test and gives effervescences with $NaHCO_3$

3) Number of moles of $LiAlH_4$ required per mole of B for the complete reaction is 2

36. How many of the following are natural polymers

a) glucose b) PVC c) Cellulose d) cis polyisoprene

e) PHBV f) DNA g) Tryptophan

37. How many of the following has (have) S-S linkage with two different oxidation state of sulphur?

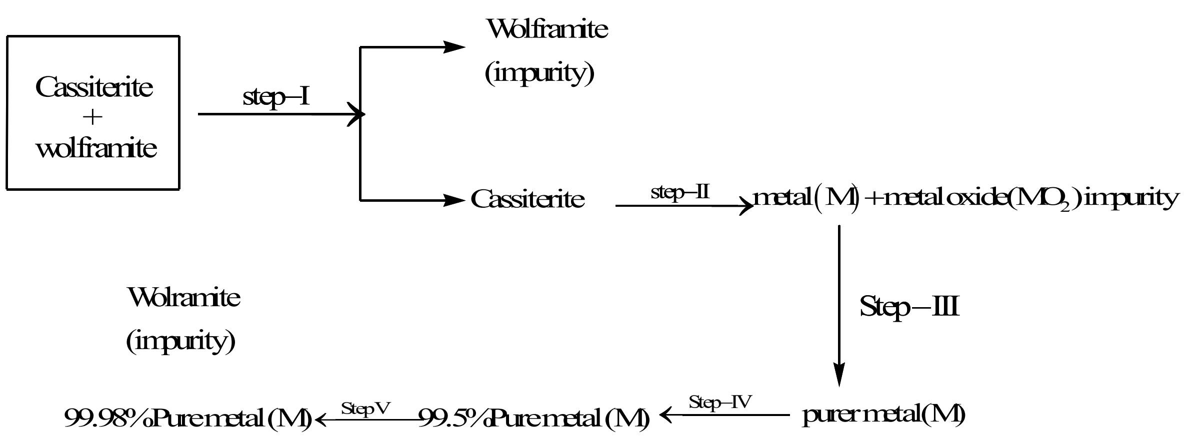
1) Pyro sulphuric acid 2) Thiosulphuric acid

3) tetrathionic acid 4) dithionic acid

5) dithionous acid 6) pyrosulphurous acid

38. Number of geometrical isomers of the complex $[Fe\,NH_3\,H_2O\,Br\,Cl\,I\,Py]$ where Br, Cl and I are at adjacent positions with respect to each other are:

39. Consider the following extraction process:



The number of correct statement(s) among the following is

1) The flow chart represents extraction of tin

2) Step-I is levigation

3) Step-II is carbon reduction

4) Step-III is poling

5) Step-IV is liquation

6) Step-V is electro refining

40. How many of the following reactions evolve a paramagnetic gas?

a) $Fe + 80\% \text{conc} \, HNO_3 \rightarrow$

b) $Cu + \text{dil.}HNO_3 \rightarrow$

c) $Zn + \text{dil} \, HNO_3 \rightarrow$

d) $Pb + \text{dil} \, HNO_3 \rightarrow$

e) $Zn + NaOH \rightarrow$

f) $(NH_4)_2 SO_4 + NaOH \rightarrow$

g) $KO_2 + H_2O \rightarrow$

MATHEMATICS **Max Marks: 60**

SECTION-1

(ONE OR MORE OPTIONS CORRECT TYPE)

This section contains 10 multiple choice equations. Each question has four choices (**A**) (**B**),(**C**) and (**D**) out of which **ONE** or **MORE THAN ONE** are correct.

Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases.

41. Which of the following can be a root of a polynomial in x of the form $9x^5 + ax^3 + b = 0$ where a and b are integers

A) -9 B) -5 C) $\frac{1}{4}$ D) $\frac{1}{3}$

42. Consider the function defined by $f(x) = e^{-x}$ on the interval $[0,10]$ let n > 1 and let $0, x_1, x_2, x_3, ... x_n$ be the numbers such that $0 = x_1 < x_2 < x_3 < ... < x_n = 10$, which of the following is greatest

A) $\sum_{j=1}^{n} f(x_j)(x_j - x_{j-1})$

B) $\sum_{j=1}^{n} f(x_{j-1})(x_j - x_{j-1})$

C) $\sum_{j=1}^{n} f\left(\frac{x_j + x_{j-1}}{2}\right)(x_j - x_{j-1})$

D) $\int_0^{10} f(x)dx$

43. Let $M_n = \{0.a_1a_2a_3......a_n / a_i = 0 \text{ or } 1, \forall 1 \le i \le n-1 \text{ and } a_n = 1\}$ be a set of decimal fractions. T_n and S_n be the number of elements and sum of elements in M_n respectively then which of the following is /are true

A) $T_n = 2^{n-1}$

B) $T_n = 2^n$

C) $S_n = \frac{2^{n-1}}{18}\left(1+\frac{8}{10^n}\right)$

D) $\lim_{n\to\infty} \frac{S_n}{T_n} = \frac{1}{18}$

44. Consider $I = \int \frac{dx}{(x^2+a^2)(x^2+b^2)}$, then which of the following is/are true?

A) For $a \ne 0, b = 0$, $I = \frac{-1}{a^2x} - \frac{1}{a^3}\tan^{-1}\left(\frac{x}{a}\right) + c$ (c – being integration constant)

B) For $a = b \ne 0$, $I = \frac{1}{2a^3}\tan^{-1}\left(\frac{x}{a}\right) + \frac{x}{2a^2(x^2+a^2)} + c$ (c – being integration constant)

C) For $a = 0, b \ne 0$, $I = \frac{-1}{b^2x^2} - \frac{1}{b^3}\tan^{-1}\left(\frac{x}{b}\right) + c$ (c – being integration constant)

D) For $a \ne 0, b \ne 0$, $I = \frac{1}{b^2-a^2}\left[\frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) - \frac{1}{b}\tan^{-1}\left(\frac{x}{b}\right)\right] + c$ (c – being integration constant)

45. Which of the following is / are true

A) tangent at a point $P(a\cos\theta, b\sin\theta)$ to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, intersects the tangent at $Q(a\cos\theta, a\sin\theta)$ to the curve $x^2+y^2=a^2$ on the x axis

B) tangent at a point $P(a\cos\theta, b\sin\theta)$ to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, intersects the tangent at $Q(a\cos\theta, a\sin\theta)$ to the curve $x^2+y^2=a^2$ not on the x axis

C) $P_1(\theta_1), P_2(\theta_2)$ are two points on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$. Given that the tangents to the ellipse at P_1, P_2 meet on the line $bx = ay$, then the value of $\theta_1+\theta_2$ can be $\frac{\pi}{2}$

D) $P_1(\theta_1), P_2(\theta_2)$ are two points on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$. Given that the tangents to the ellipse at P_1, P_2 meet on the line $bx = ay$, then the value of $\theta_1+\theta_2$ can be $\frac{5\pi}{2}$

46. Which of the following is/are true ?

A) $\sum_{v=0}^{n} \frac{(2n)!}{(v!)^2 ((n-v)!)^2} = \left({}^{2n}C_n\right)^2$

B) $\sum_{k=1}^{n} \left({}^{n}C_k\right) \frac{(-1)^{k-1}}{k} = \sum_{k=1}^{n} \frac{1}{k}$

C) $\sum_{k=0}^{n} \left({}^{k+r-1}C_{r-1}\right) = {}^{n+r}\ C_r$

D) $\sum_{k=0}^{n} (-1)^k \left({}^{a}C_k\right) = (-1)^n \left({}^{a-1}C_n\right)$

47. Let $f:[0, 1] \to (0, \infty)$ be continuous function such that $\int_0^1 f(x)dx = 1$ also $I = \left(\int_0^1 \sqrt[3]{f(x)}\,dx\right)\left(\int_0^1 \sqrt[5]{f(x)}\ dx\right)\left(\int_0^1 \sqrt[7]{f(x)}\ dx\right)$, then which of the following can be true?

A) $I \in \left[\frac{1}{2}, \frac{3}{4}\right]$ B) $I \in \left[\frac{1}{4}, \frac{1}{2}\right]$ C) $\left[\frac{3}{2}, 2\right]$ D) $\left[\frac{5}{4}, \frac{3}{2}\right]$

48. The point $P(a\cos\theta, b\sin\theta)$ on the ellipse corresponds to the point $A(a\cos\theta, a\sin\theta)$ on the auxillary circle $x^2 + y^2 = a^2$ and OP meets the auxiliary circle at $B(a\cos\phi, a\sin\phi)$ (O- origin) then which of the following is / are true

A) the tangent drawn to the ellipse at $Q(a\cos\phi, b\sin\phi)$ corresponds to B is perpendicular to OA

B) the tangent drawn to the ellipse at $Q(a\cos\phi, b\sin\phi)$ corresponds to B, is inclined at 60^0 to OA

C)the tangent drawn to the ellipse at $Q(a\cos\phi, b\sin\phi)$ corresponds to B, meets OA at T, then OP = OT

D) the tangent drawn to the ellipse at $Q(a\cos\phi, b\sin\phi)$ corresponds to B, meets OA at T, then OP = $\frac{2}{3}$OT

49. A bag contains N_1 white and N_2 black balls $(N_1 \neq 0, N_2 \neq 0)$. When two balls are randomly drawn, the probability that both will be white is $\frac{1}{2}$ then which of the following is/are true?

A) minimum value of N_1 is 4

B) minimum value of N_1 is 3

C) minimum value of N_1 is 12 when N_2 is even

D) minimum value of N_1 is 15 when N_2 is even

50. Let $X=\sum_{r=0}^{n}\left(\frac{(-1)^r}{r+1}\right)$, then $|X-\ln 2|$, will be

A) greater than $\frac{1}{2(n+1)}$

B) greater than $\frac{1}{2(n+2)}$

C) less than $\frac{1}{2(n+1)}$

D) less than $\frac{1}{2(n+2)}$

SECTION-II

INTEGER TYPE

This section contains 10 questions. The answer to each question is a **single digit integer, ranging** from 0 to 9 (both inclusive).

Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases.

51. If differential equation of the family given by $e^{2y}+2cxe^{y}+c^2=0$ (c – is a parameter) is $(1-x^2)\left(\frac{dy}{dx}\right)^2+\lambda=0$, where $\lambda=$_______

52. The number of ordered triads of complex numbers (x,y,z) which satisfy the condition

$x(x-y)(x-z)=y(y-x)(y-z)=z(z-x)(z-y)=3$ is…

53. If $L=\underset{n\to\infty}{lt}\frac{1}{n}\left\{\sum_{r=1}^{n}\left(\sin\left(\frac{r\pi}{2n}\right)-r\right)\left(\sin\left(\frac{r\pi}{2n}\right)+r\right)+\int_0^n\left(x^2+x+\frac{1}{6}\right)dx\right\}$ then 2L =

54. The number of ways in which 5 similar ice creams and 3 different flavor pizzas can be distributed to 7 children so that each child gets atleast one item is a four digit number of the form "abcd", where a+b = ________

55. If $\lim_{n\to\infty}\frac{1}{n}\sum_{k=1}^{n}k\ln\left(\frac{n^2+(k-1)^2}{n^2+k^2}\right)$ exists and is equal to L. Find the absolute value of [L]……

[Note : [y] denotes greatest integer less than or equals to y.]

56. Coefficient of x^{13} is the expansion of $(1-x)^5\left(1+x+x^2+x^3\right)^4\left(1+x^{13}\right)^8$ is k then $k-6=$ ______

57. Consider the circle $|Z-5i|=3$ and two points Z_1 *and* Z_2 on it such that $|Z_1|<|Z_2|$ *and* $\arg(Z_1)=\arg(Z_2)=\frac{\pi}{3}$ A tangent is drawn at Z_2 to the circle, which cuts the real axis at Z_3, then $[|Z_3|]=$ ----------

[Note : [y] denotes greatest integer less than or equals to y.]

58. There are 6 balls of each of the four colours: red, white, yellow and black in a bag (balls of same colour are identical). Let n be the total number of different ways of drawing 6 balls one by one without replacement such that no two consecutive balls are of the same colour and no colour is missing in the draw. Find the non–zero digit in n. …

59. At how many points in the xy – plane do the graphs of $y=x^{12}$ and $y=2^x$ intersect ..

60. If radii of three concentric circles are related as $r_1(r_2+r_3)+r_2(r_3+r_1)+r_3(r_1+r_2)=118$, $r_1r_2(r_1+r_2)+r_2r_3(r_2+r_3)+r_3r_1(r_1+r_3)=616$ and $\sum\left(\frac{r_1^2+r_2^2}{r_1r_2}\right)=\frac{44}{5}$ then difference between the radii of the largest and smallest circle is………

KEY SHEET

1	**BD**	2	**ABC**	3	**BD**	4	**BC**	5	**BC**
6	**ACD**	7	**AC**	8	**AC**	9	**ACD**	10	**AC**
11	**2**	12	**2**	13	**5**	14	**6**	15	**5**
16	**5**	17	**9**	18	**8**	19	**6**	20	**5**
21	**ABC**	22	**ABD**	23	**ABC**	24	**ABCD**	25	**BD**
26	**ABD**	27	**ABCD**	28	**ABCD**	29	**BCD**	30	**BC**
31	**9**	32	**7**	33	**3**	34	**6**	35	**2**
36	**3**	37	**3**	38	**6**	39	**5**	40	**3**
41	**ABD**	42	**B**	43	**ACD**	44	**ABD**	45	**ACD**
46	**ABCD**	47	**AB**	48	**AC**	49	**BD**	50	**BC**
51	**1**	52	**6**	53	**1**	54	**6**	55	**1**
56	**6**	57	**3**	58	**6**	59	**3**	60	**5**